



Pritam Mitra

Nationality: Indian

Date of birth: 02/02/1997

Place of birth: Itahar, India

Gender: Male

Phone number: (+91) 8906009315

Email address: pritam.mitra97@gmail.com, pritam22phd@cup.ac.in

Home: Village-Chalunia, Post Office & Police Station- Itahar, District- Uttar Dinajpur, State- West Bengal, PIN-733128 (india)

Work: Central University of Punjab, Vill- Ghuddha, Post Office- Sangat, Police Station & District-Bhatinda, State- Punjab, PIN- 151401 (India)

Research Experience

Department of Geography, School of Environment and Earth Science, **Central University of Punjab (Under Ministry of Education, Government of India)** India [26/04/2023 – Current]

Address: Vill- Ghuddha, Post Office- Sangat, Police Station & District-Bhatinda, State- Punjab, PIN- 151401, Country- India | Website: https://cup.edu.in/evs_sciences.php/centre_geography.php | Level in EQF: Level 8

Research Scholar (Ph.D.)

Role & Responsibilities:

- Research under the supervision of a guide/supervisor.
- Assist in teaching/mentoring.

Education

Academic courses: -

Doctor of Philosophy (Ph.D.)

Department of Geography, School of Environment and Earth Science, **Central University of Punjab (Under Ministry of Education, Government of India)**, India [26/04/2023 – Current] Level in EQF: Level 8

Address: Vill- Ghuddha, Post Office- Sangat, Police Station & District-Bhatinda, State- Punjab, PIN- 151401, Country- India | Website: https://cup.edu.in/evs_sciences.php/centre_geography.php

Under the supervision of: **Dr. Ir. Sk. Mustak, Assistant Professor**, Department of Geography, School of Environment and Earth Science, Central University of Punjab | Email id- mustak.sk5@gmail.com

Completed Ph.D. course work with CGPA: 8.00

Master of Science (M.Sc)

University of Calcutta (A public state university, Government of West Bengal, India) [01/07/2017 – 29/11/2019]

Address: Tarak Nath Palit Siksha Prangan (Ballygunge Campus), Post Office- Ballygunge, Police Station Ballygunge, District- Kolkata, State- West Bengal, PIN- 700019, (India) | Contacts: <https://www.caluniv.ac.in/admin/officer.html> | Website: <https://www.caluniv.ac.in/>

Field(s) of study: Geography | University: University of Calcutta | **Final grade: (First Class) 64.90 percent** | Level in EQF: Level 7

Bachelor of Science (B.SC Honours)

Asutosh College (Under University of Calcutta, A public state university, Government of West Bengal, India) [25/06/2014 – 23/06/2017]

Address: 92, Shyamaprasad Mukherjee Road, Post Office- Kalighat, Police Station- Bhowanipore, District Kolkata, State- West Bengal, PIN- 700026, (India) | Email id: mail@asutoshcollege.in | Website: <https://asutoshcollege.in/new-web/>

Field(s) of study: Geography Honours with Economics & Sociology | **University:** University of Calcutta | **Final grade: (First Class) 68.12 percent** | Level in EQF: Level 6

Senior School Certificate Examination

Jawahar Navodaya Vidyalaya, North 24 Parganas, (Under Navodaya Vidyalaya Samiti, An Autonomous Body Under Ministry of Education, Government Of India) [04/08/2012 – 28/05/2014]

Address: Banipur, Post Office- Banipur, Police Station- Habra, District- North 24 Parganas, State- West Bengal, PIN- 743232, (India) | Email id: jnvb24pgs@gmail.com | Website: https://navodaya.gov.in/nvs/nvs-school/NORTH24PARGANAS/en/about_us/About-JNV/

Field(s) of study: English Core, Bengali, History, Geography, Economics | **Board:** Central Board of Secondary Education | **Final grade: (First Class) 91.4 percent** | Level in EQF: Level 4

Secondary School Examination

Jawahar Navodaya Vidyalaya, Uttar Dinajpur, (Under Navodaya Vidyalaya Samiti, An Autonomous Body Under Ministry of Education, Government Of India) [01/08/2007 – 24/05/2012]

Address: Teesta Colony, Post Office- Jagadishpur, Police Station- Dalkhola, District- Uttar Dinajpur, State West Bengal, PIN- 733201, (India) | Email id: jnvdalkhola@gmail.com | Website: https://navodaya.gov.in/nvs/nvs-school/UttarDinajpur/en/about_us/About-JNV/

Field(s) of study: English Language & Literature, Bengali, Mathematics, Science, Social Science | **Board:** Central Board of Secondary Education | **Final grade: (First Class) 91.2 percent** | Level in EQF: Level 3

Professional courses: -

Bachelor of Education (B.Ed.)

Nazrul Islam College of Education (Under West Bengal University of Teachers' Training, Education Planning and Administration (Renamed as Baba Saheb Ambedkar Education University), A public state university, Government of West Bengal, India) [01/12/2019 – 10/08/2021]

Address: Village & Post Office- Bara Andulia, Police Station- Chapra, District- Nadia, State- West Bengal, PIN- 741124, (India) | Email id: nicollegeofeducation@gmail.com | Website: <https://www.nicollegeofeducation.org/contact.html>

Field(s) of study: Geography as method subject | **University:** West Bengal University of Teachers' Training, Education Planning and Administration | **Final grade: (First Class) 83.65 percent** | Level in EQF: Level 6 | **Additional achievement:** Successfully completed a 14-week school internship and a 2-week community engagement programme as per NCTE rule, which involved classroom observation, lesson planning, supervised teaching practice, student evaluation, and participation in community-based educational activities.

Technical / vocational courses: -

Diploma in Computer Application (DCA)

Indira Gandhi National Youth (Government of India Approved) [15/01/2019 – 22/02/2020]

Address: Ranaghat IGNY, Post Office & Police Station- Ranaghat, District- Nadia, State- West Bengal, PIN741201, (India) | Email id: ignyindia@gmail.com | Website: <http://www.ignyindia.org/NewIgny/index.html>

Field(s) of study: Computer application | Board: Indira Gandhi National Youth | **Final grade: (First Class) 76.2 percent** | Level in EQF: Level 4

Language

Mother tongue(s): Bengali

Other language(s):

English

LISTENING C2 READING C2 WRITING C2

SPOKEN PRODUCTION C2 SPOKEN INTERACTION C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

Hindi

LISTENING C2 READING C2 WRITING C2

SPOKEN PRODUCTION C2 SPOKEN INTERACTION C2

Research areas

 Flood Risk & Hazard Assessment – Flood mapping, vulnerability analysis, flood modeling using GIS, Remote Sensing, SAR and Landsat data.

 Urban Thermal Studies – Urban Heat Island (UHI) and Urban Cold Island (UCI) analysis using Landsat-derived LST, spatial mapping, and trend assessment.

 Climate Change & Variability – Climate data analysis, trend detection, impact assessment using historical and projected datasets (IMD, CMIP6), including SAR-derived environmental monitoring.

 Land Subsidence & Geohazards – Monitoring land subsidence, urban settlement stability, and integrated hazard mapping using SAR Interferometry (InSAR) and geospatial techniques.

 Disaster Risk Management – Hazard assessment, vulnerability mapping, and mitigation strategies for climate-induced disasters.

 Socio-Economic & Social Inequality Studies – Analysis of social vulnerability, inequality mapping, and integration of demographic and environmental data.

 Geospatial & Spatial Statistics Applications – Frequency Ratio, Weight of Evidence, Certainty Factor, Logistic Regression, and multi-criteria decision-making for hazard and vulnerability modeling.

 Applied AI & Machine Learning in Environmental Studies – Predictive modeling for flood, groundwater, hazard, and thermal mapping using ensemble algorithms (Random Forest, SVM, Boosting, etc.).

Merits, Awards & Scholarships (14)

National:-

- 2025. **Inspirational Award** for *Outstanding Paper* in the **Prof. B. Hema Malini Young Geographer Award Session**, organized by the **Deccan Geographical Society of India (DGSi)**; awarded a **cash prize of INR 3,000**.
- 2025. **Certificate of Excellence Award** for *Outstanding Paper* at the 7th Regional Science & Technology Congress, organized by the **Department of Science & Technology (DST), Government of West Bengal, India**; awarded a **cash prize of INR 10,000**.
- 2022. 24th West Bengal State Eligibility Test (WBSET), West Bengal College Service Commission (WBCSC), Government of West Bengal, India.
- 2022. **National Eligibility Test – Junior Research Fellowship (NET-JRF)**, University Grants Commission (UGC), Government of India, New Delhi – Qualified through this highly competitive All-India entrance examination (December 2021 & January 2022 merged cycle); **fellowship of INR 25,65,000 awarded for pursuing Ph.D.**
- 2022. National Eligibility Test (NET), University Grants Commission (UGC), Government of India, New Delhi – Qualified in the December 2020 & June 2021 merged cycle.
- 2019. **Swami Vivekananda Merit-cum-Means Scholarship**, Government of West Bengal, India awarded to *meritorious M.Sc. students*; **scholarship amount INR 30,000**.
- 2018. **Swami Vivekananda Merit-cum-Means Scholarship**, Government of West Bengal, India awarded to *meritorious M.Sc. students*; **scholarship amount INR 30,000**.
- 2017. **Certificate of Merit** awarded by the University of Calcutta, India for securing **Sixth Rank in the Bachelor of Science (Honours) Examination**.
- 2016. **Central Sector Scheme of Scholarship (CSSS)**, Government of India – *Awarded to meritorious college and university students*; **scholarship amount INR 10,000**.
- 2015. **Central Sector Scheme of Scholarship (CSSS)**, Government of India – *Awarded to meritorious college and university students*; **scholarship amount INR 10,000**.
- 2014. **Letter of Appreciation** by the **Hon'ble HRD Minister, Government of India, Smt. Smriti Zubin Irani**, for *outstanding performance in CBSE Class XII All-India Senior School Certificate Examination*.
- 2014. **Certificate of Excellence award** by CBSE, Government of India, for participation in Abhinandan (a programme honouring CBSE Class XII meritorious students) at New Delhi, along with invitation to the Independence Day Celebration at Red Fort.
- 2014. **Certificate of Merit by CBSE**, Government of India, for *outstanding academic performance and securing position among the top 0.1% of successful candidates in Geography* in the All-India Senior School Certificate Examination.
- 2007. **Selection for Jawahar Navodaya Vidyalaya (JNV), Ministry of Education, Government of India** – Merit-based admission through national entrance examination; **received free residential education (Class VI–XII)** at this CBSE-affiliated school for meritorious rural students, **with all expenses fully**

Publications (Total-3)

1. **Mitra, P.**, Mustak, S., Pradhan, B., Pulak Barman, ., Prosenjit Barman, ., Amol, ., & Garg, K. (2026). SAR-based flood detection and different ensemble boosting techniques for multi-factor flood susceptibility modelling: a case study in Uttar Dinajpur, West Bengal, India. *Natural Hazards* 2026 122:10, 122(10), 438-. <https://doi.org/10.1007/S11069-026-08171-0>
2. **Mitra, P.**, Mustak, S., Barman, P., Garg, A. K., Barman, P., & Pradhan, B. (2026). A novel Cold Island Core Network (CICN) approach for urban heat island mitigation using machine learning and GIS. *Physics and Chemistry of the Earth*, 144(P1), 104405. <https://doi.org/10.1016/j.pce.2026.104405>

3. Arpita, A., Mustak, S., **Mitra, P.**, Barman, P., Barman, P., Garg, A. K., & Singh, D. (2025). Integrated flood risk-based land use mapping using machine learning and Geospatial technologies: a case study of Baleswar CD block, India. *Discover Cities* 2025 2:1, 2(1), 128-. <https://doi.org/10.1007/s44327-025-00173-0>.

Articles under review (7) :-

1. **Mitra, P.**, Mustak, S.K., Pradhan, B., Barman, P., Barman, P., Garg, A.K. (2025). *SAR-based flood detection and different ensemble boosting techniques for multi-factor flood susceptibility modelling: a case study in Uttar Dinajpur, West Bengal, India*, **Natural Hazard** (under 2nd revision) (Date of submission:- May, 2025).
2. Mattam, R.B., Barman, P., Mustak, S.K., Barman, P., **Mitra, P.** (2025). *Spatio-temporal Land Use Land Cover Alteration Impact on Urban Heat Island Amplification in Hyderabad City Using Geospatial Technologies*, **Human Settlements and Sustainability** (under 1st revision) (Date of submission:- July, 2025).
3. Sarkar, A., Barman, P., Mustak, S.K., Alam, S., Pradhan, B., **Mitra, P.** (2025). *Mapping the Evolving Landscape and Future Trend of Artificial Intelligence in Global Tourism Research in 21st Century Using Bibliometric Analysis*, **Geoheritage Journal** (under 1st revision) (Date of submission:- August, 2025).
4. **Mitra, P.**, Mustak, S.K., Barman, P., Garg, A.K., Barman, P., (2025). *Assessing the impact of urbanization and land use land cover dynamics on the Ecological Health of the Kolkata Metropolitan Area*, **Ecological Modelling** (under 1st revision) (Date of submission:- August, 2025).
5. **Mitra, P.**, Mustak, S.K., Barman, P., Garg, A.K., Barman, P., (2025). *From Thermal Stress to Climate Relief: Building cold island network to alleviate urban heat island effect using machine learning and GIS-based hybrid models*, **Urban Climate** (under 1st revision) (Date of submission:- September, 2025).
6. **Pal. S., Mitra, P.**, Mustak, S.K., Bika, S., Pradhan, B., (2025). *Global Research Trends of STEAM Education: A Bibliometric Analysis*, **Educational Science** (under 1st revision) (Date of submission:- September, 2025).
7. **Mitra, P.**, Mustak, S.K., Pradhan, B. *Prediction of Future Peak Vegetation Dynamics across India Using XAI-Based Machine Learning Approaches and CMIP6 Projections*, **Natural Hazard** (under revision) (Date of submission:- December, 2025).

Research & Technical skills

 **Remote Sensing & GIS** – Satellite image processing, LULC classification, DEM analysis, flood & groundwater mapping; experienced with SAR, Landsat, Sentinel, and Planet Scope datasets.

 **Geospatial Software** – QGIS, ArcGIS, ERDAS Imagine, ENVI, SNAP, Conefor, Guddos, Google Earth Engine.

 **Applied AI & Machine Learning** – Supervised & unsupervised learning, ensemble methods (Random Forest, SVM, AdaBoost, Gradient Boosting, XGBoost, LightGBM, CatBoost), predictive environmental & hazard modeling.

 **Statistical Analysis & Programming** – SPSS, R, Python (pandas, scikit-learn, matplotlib); raster/vector data handling and automation of geospatial workflows.

 **Spatial Statistics** – Logistic Regression, Frequency Ratio, Weight of Evidence (WOE), Certainty Factor.

 **Climate & Hydrological Data Analysis** – IMD datasets, CMIP6 climate projections, Mann-Kendall, Sen's Slope.

 **Digital Cartography & Visualization** – Thematic map design, uncertainty mapping, geospatial data visualization.

 **Surveying & Instruments** – Prismatic Compass, Theodolite, Total Station, Dumpy Level, GPS-based mapping.

 **Field Survey** – Socio-economic surveys, ground-truthing, primary data collection.

Conference / seminar presentation (Total-8)

International conferences / seminar:-

1. Mitra, P., Mustak, S., (2025). From Space to Cool Corridors: A Machine Learning and GIS Approach to Urban Cold Islands network, **6th Asian Conference on Geography** during 6th to 8th November, hosted by the Department of Geography, Jamia Millia Islamia, New Delhi, India in Collaboration with Asian Geographical Association (AGA) (Oral presentation).
2. Mitra, P., (2025). Spatio-temporal mapping and monitoring of land use and land cover dynamics and assessing its impact on eco-environmental quality using earth observation: A machine learning based geospatial approach, **Professor B. Hema Malini Young Geographer Award Session, 2025** by The Deccan Geographical Society on 22nd February, 2025, hosted by the Department of Geography, Central University of South Bihar, Bihar, India (Oral Presentation).
3. Mitra, P., Mustak, S., (2025). Machine learning based flood susceptibility mapping in Uttar Dinajpur district of West Bengal, India using SAR data and earth observation, **XIX DGSi International Geography Conference** during 22nd to 24th February, 2025, hosted by the Department of Geography, Central University of South Bihar, Bihar, India (Oral Presentation).
4. Mitra, P., Mustak, S., Barman, P., (2023). Monitoring ecological crisis of Kolkata Metropolitan Area using spatial technology, **43rd Indian National Cartographic Association (INCA) International Congress** during 6th to 8th November, 2023, hosted by the Regional Remote Sensing Centre-West, National Remote Sensing Centre, Indian Space Research Organization, Jodhpur, Rajasthan, India (Oral Presentation).
5. Mitra, P., Mustak, S., (2023). Assessing the impact of urbanisation on wetland ecosystem service using InVEST model and earth observation: A case study of Kolkata Metropolitan Area, **International seminar on Recent Advancement in Geographical Studies** during 1st to 2nd September, 2023, hosted by the Department of Geography, Rampurhat College, Birbhum, West Bengal, India (Oral Presentation).

National conferences / seminar:-

1. Mitra, P., Mustak, S., (2025). Machine learning based land use land cover mapping and monitoring its impact on urban ecological health using earth observation: a case study of Kolkata Metropolitan Area, **7th Regional Science & Technology Congress** during 7th to 8th January, 2025, jointly hosted by the Malda College, Malda and Department of Science and Technology and Biotechnology, Government of West Bengal, India (Oral Presentation).
2. Mitra, P., Mustak, S., (2025). Machine learning based land use land cover mapping and monitoring its impact on urban ecological health using earth observation: a case study of Kolkata Metropolitan Area, **32nd West Bengal State Science & Technology Congress** during 28th February to 2nd March, 2025, hosted by the Department of Science and Technology and Biotechnology, Government of West Bengal, India (Oral Presentation).
3. Paul S., Mitra, P., Mustak, S., (2025). A bibliometric analysis on the global research trends of STEAM education, **ICSSR sponsored two days National Seminar** during 17th to 18th March, 2025, hosted by the Department of Education, Central University of Punjab, Punjab India (Oral Presentation).

Cutting-edge Training & Workshops (Total-9)

1. Workshop on “Groundwater quality issues and challenges in Punjab, India” DST-SERB (ANRF) sponsored, held offline during 16th to 17th February 2025, hosted by the Department of Environmental Science and Technology, Central University of Punjab, India.
2. Training on “**Artificial Intelligence in Remote Sensing and Earth Observation**”, AICTE sponsored, held offline during 20th to 25th January 2025, hosted by the Sikkim Manipal institute of technology, Sikkim, India.
3. Workshop on “**AI Research with MATLAB & SIMULINK**”, CUPB-sponsored, held offline during 26th to 27th November, 2024 at the Computer Centre, Central University of Punjab, Punjab, India.
4. Training on “**NEP 2020 Orientation & Sanitization**”, under Malaviya Mission Teacher Training Programme, UGC sponsored, held online during 20th to 29th May 2024, hosted by the Central University of Punjab, Punjab, India.

5. Workshop on “*Finishing School*”, CUPB-sponsored, held offline during 10th to 13th April 2024 at the Central University of Punjab, Punjab, India.
6. Workshop on “*Application of statistics and machine learning in environmental research*”, DST SERB-sponsored, held offline during 4th to 10th March 2024 at the Indian Statistical Institute North-East Centre, Tezpur, Assam, India.
7. Workshop on “*Research Methodological Course*”, ICSSR-sponsored, held offline during 5th to 14th February, 2024 at the Department of Geography, Central University of Punjab, Punjab, India.
8. Workshop on “*Applications of Machine learning in Urban Studies*”, IIRS sponsored, held online during 5th to 30th July 2023, hosted by Indian Institute of Remote Sensing, Dehradun, India.
9. Workshop on “*Artificial Intelligence, IoT & Sensors for Agriculture, Forestry and Environmental Conservation*”, DST SERB-sponsored, held offline during 24th to 30th July 2023 at the Department of

Research profiles

Google Scholar , ORCID , ResearchGate 

Declaration

I, hereby declare that all the information furnished above, are true to the best of my knowledge and belief.

Date: 10/04/2026

Place: Bhatinda, India

(Pritam Mitra)

References

1. Dr. ir. Sk Mustak

MS (ITC, Netherland). M. Phil. Ph.D.

[Earth observation and urban Geoinformatics Lab]

Assistant Professor,

Department of Geography, School of Environment and Earth Sciences.

Central University of Punjab (Ministry of Education, Governmet of India)

Contact: Email  **Research Profile:** Google Scholar , ORCID , ResearchGate 